Mohamad Moussa

Electrical Engineer | Power Systems

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Available immediately | Transferable dependent iqama (Saudi Arabia)

SUMMARY

Electrical engineer (KFUPM, First Honors) specializing in power system studies: load flow, contingency analysis, and system planning. Led team projects from analysis to remedial design; adds Power BI, SQL, and Python for engineering data. Seeking graduate and junior roles in power systems, studies, testing, and project engineering in the GCC and Europe.

SKILLS

- **Power systems:** Load flow (fast decoupled), N-1 contingency analysis, system planning, reactive power compensation, transformer tap control, Y-bus modeling
- **Software:** PowerWorld, MATLAB, Python, SQL, Power BI, DAX, Excel
- **Languages:** Arabic (native), English (fluent)

PROJECTS

IEEE 14-Bus Power System Study | PowerWorld, MATLAB | May 2023 | Team lead (3 members)

- Led the electrical analysis: fast-decoupled power flow, transformer tap sensitivity, and N-1 contingency analysis, with the Y-bus verified against MATLAB code.
- Under a 90% load increase, found 8 overloaded lines and restored limits using parallel lines, load shedding, capacitor banks, and voltage setpoint changes.

Automated Online Electric Motor Testing (Capstone) | KFUPM | 2023–2024 | Team of 5 (EE, ISE, COE)

- Built an automated online motor-testing solution with real-time performance monitoring for power utilities, benchmarked against commercial products using QFD.

Load Forecasting of Saudi National Demand | 2023 | Team lead (3 members)

- Compared seven forecasting methods on hourly demand data; weighted hourly moving average reached 1.65% MAPD and a same-weekday naïve baseline 0.69%.

Solar PV Power Forecasting with AI | Python | 2023 | Co-author

- Benchmarked seven models (LSTM, GRU, ConvLSTM, N-BEATS, Transformer, others) on 4,214 samples; best R^2 of 0.81 (N-BEATS).

Power BI Dashboards (Sales Intelligence, Customer Churn) | Power BI, DAX | Jun 2026

- Built two five-page dashboards; found 20.37% overall churn, with Germany (32.44%) and ages 46–60 (51.12%) at highest risk.

EXPERIENCE

Research Assistant, SURE Program | KFUPM, Dhahran | Summer 2023

- Tested and validated iterative error-cancellation techniques for stable dynamical systems, comparing manual and automated approaches; documented results in a 46-page technical report.

EDUCATION

B.Sc. Electrical Engineering, First Honors | KFUPM, Dhahran | Jan 2024 | GPA 3.82/4.00

Coursework: Power System Analysis, Power Transmission & Distribution, Smart Grids, Power System Planning & Operation, Senior Design Project

CERTIFICATIONS

Power BI (Harnessing the Power of Data; ETL in Power BI) | Data Analysis using Python | Preparing Data with Excel | Project Management Fundamentals